

Lab 1:

The screenshot shows the pgAdmin 4 interface with a query executed against the 'dvdrental/postgres@PostgreSQL 18' database. The query is a SELECT statement that joins 'pg_class' with 'pg_namespace' to retrieve table and index sizes. The results are displayed in a table with columns: object_name, data_size, index_size, and total_size. The data is sorted by total_size in descending order.

```
SELECT
    relname AS object_name,
    pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
    pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
    pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
FROM pg_class fed
JOIN pg_namespace n ON n.oid = fed.relnamespace
WHERE fed.relkind = 'r' AND n.nspname = 'public'
ORDER BY pg_total_relation_size(fed.oid) DESC;
```

	object_name	data_size	index_size	total_size
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	film_category	72 kB	40 kB	112 kB
9	city	64 kB	48 kB	112 kB
10	actor	40 kB	32 kB	72 kB

Total rows: 17 Query complete 00:00:00.689 CRLF Ln 2, Col 13

The index-heavy tables are rental, payment, customer, store, language, category, country, rental_log and secure_orders, where the index size is $\geq$ the table data size

Lab 2:

The screenshot shows the pgAdmin 4 interface with a query executed against the 'dvdrental/postgres@PostgreSQL 18' database. The query creates an index on the 'payment' table and then calculates its size in bytes and pages consumed.

```
SELECT pg_size_pretty(pg_indexes_size('payment'));
CREATE INDEX idx_payment_date_long ON payment (payment_date, amount, customer_id);
SELECT pg_size_pretty(pg_indexes_size('payment'));

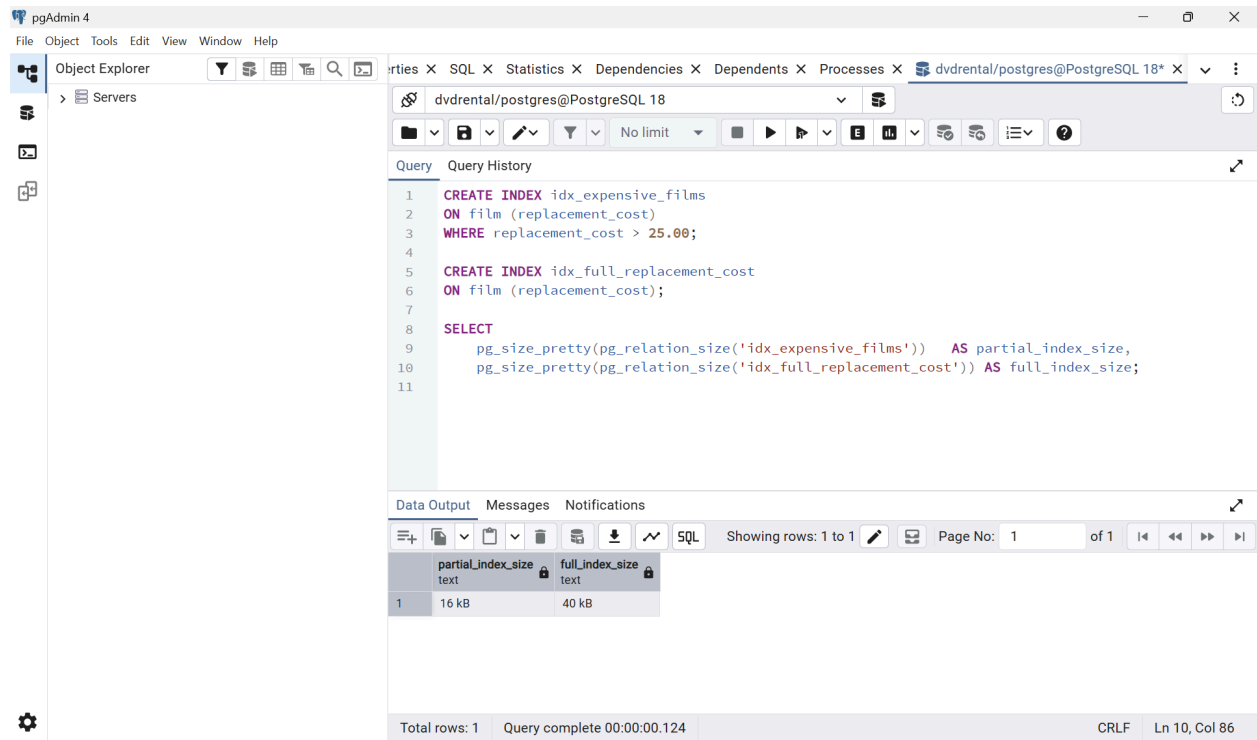
SELECT
    pg_relation_size('idx_payment_date_long') AS size_bytes,
    pg_relation_size('idx_payment_date_long') / 8192 AS pages_consumed;
```

	size_bytes	pages_consumed
1	581632	71

Total rows: 1 Query complete 00:00:00.162 CRLF Ln 7, Col 72

There are 71 8KB pages that this new index consumed

Lab 3:



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query creates two indexes on the 'film' table: a partial index 'idx_expensive_films' for rows where 'replacement_cost' is greater than 25.00, and a full index 'idx_full_replacement_cost'. It then uses 'pg_size_pretty' to show the sizes of these indexes.

```
1 CREATE INDEX idx_expensive_films
2 ON film (replacement_cost)
3 WHERE replacement_cost > 25.00;
4
5 CREATE INDEX idx_full_replacement_cost
6 ON film (replacement_cost);
7
8 SELECT
9     pg_size_pretty(pg_relation_size('idx_expensive_films')) AS partial_index_size,
10    pg_size_pretty(pg_relation_size('idx_full_replacement_cost')) AS full_index_size;
11
```

The results are displayed in a table with two columns: 'partial_index_size' and 'full_index_size'. The first row shows the partial index size as 16 kB and the full index size as 40 kB.

	partial_index_size	full_index_size
1	16 kB	40 kB

At the bottom of the interface, it indicates 'Total rows: 1' and 'Query complete 00:00:00.124'.

The partial index (idx_expensive_films) is 16 kB, while the hypothetical full index (idx_full_replacement_cost) is 40 kB